GAUTAM SINGH

Software Engineer — backend, data, and ML systems

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EXPERIENCE

Machine Learning Intern Aug 2026 - Present
Sphere Global (automated vehicle damage inspection) *Remote*

- Build and maintain the Python tool the labeling team uses every day to prepare training data for an automated vehicle damage inspection system, running on one ordinary laptop with no GPU so customer photos never leave the site.
- Created a generator that turns clean vehicle photos into ~12,000 realistic damaged ones in NumPy and OpenCV, each labeled automatically to the pixel in YOLO format, removing the need to find, photograph, and hand-mark damaged cars.
- Steered the team off a costly dead end by benchmarking SDXL and FLUX, two leading AI image generators, against the in-house generator, which reproduced real damage more accurately, faster, and at lower cost.
- Found and fixed a flaw that had mislabeled 39% of training images, where the tool outlined the floor and walls instead of the car, by making a YOLO detector confirm the vehicle before SAM 2.1 traced it.

ServiceNow Developer & QA Specialist Sep 2025 - Dec 2025
PAMT Consulting (now Ateko, a Bell Canada company) *Remote*

- Reduced new-hire onboarding handling time by ~30% by replacing a manual, error-prone process with an automated workflow built in ServiceNow App Engine Studio and Flow Designer.

ServiceNow Developer & QA Specialist Sep 2024 - Dec 2024
PAMT Consulting *Remote*

- Designed and ran test cases across development, test, and production environments before each release.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Rust, Java, C++
Backend & Frontend: FastAPI, Django, Node.js, React, REST APIs, Redis/RQ, Flutter
Cloud & DevOps: AWS (Lambda, DynamoDB, CloudFront), Azure, Docker, Terraform, Linux, GitHub Actions CI/CD, Cloudflare
Data & ML: PostgreSQL, MongoDB, Redis, Databricks, PySpark, PyTorch, scikit-learn, Pandas, Computer Vision, OpenCV, YOLO

PROJECTS

Distill *2026*
Python, PyTorch, Qwen2.5-VL, Transformers, Pydantic, FastAPI, Docker, GitHub Actions

- Lifted delivered accuracy from 80% to 97% on a local vision-language model that reads invoices and receipts into typed JSON, scoring confidence per field by self-consistency sampling and routing what the LLM does not trust to a human.
- Kept the published numbers honest with 175 tests and a CI gate that fails the build when a metric moves.

FitFileSize | Live | GitHub *2026*
Python, FastAPI, Redis/RQ, React, TypeScript, Docker, Cloudflare Tunnel, Ghostscript, MCP

- Solo-built and launched a web app, public API, and MCP server that compress PDFs and images (7 formats, including iPhone HEIC) to fit an upload limit or a form's exact rules, such as visa photos and exam uploads.
- Designed an async job pipeline (FastAPI, Redis/RQ workers, Server-Sent Events) with format-specific logic behind one strategy interface, self-hosted with Docker on a VPS behind a Cloudflare Tunnel with deploy-on-merge.
- Halved the renders per compression with a guided search that predicts the right setting from sizes already measured, taking a 5 MB photo from 9.2 s to 4.2 s on production with byte-identical output, behind ~350 tests across three CI jobs.
- Hardened the deployment from failure drills, so a watchdog restarts any container the health endpoint reports hung and a deploy gives workers two minutes to finish where Docker allowed ten seconds.

Cairn | Live | GitHub *2026*
Rust, AWS Lambda, DynamoDB, CloudFront, API Gateway, Terraform

- Cut warm response to 5.1 ms and cold starts to 184 ms with a Rust binary on ARM64 AWS Lambda, doing one DynamoDB write per event, deferring every dimension derivation, and hashing visitors under a daily-rotating key instead of setting a cookie.
- Deployed the stack as 41 Terraform managed resources running live across three sites at zero steady-state cost.

The Almanac (Capstone) | GitHub *2025*
Python, Azure Data Factory, Databricks, PySpark, Azure SQL, PyTorch, Django REST, Flutter

- Improved data load performance by ~35% by redesigning the storage layer of an automated market-data pipeline behind a stock-tracking web and mobile app, shipped with a team over eight months.

OPEN SOURCE

actualbudget/actual | PR #8603 *Merged 2026*

- Contributed a merged bug fix to a widely used open-source personal finance app, returning a proper error when a password change is attempted with no password method set.

EDUCATION

Bachelor of Computer Science (Honors) Sep 2020 - Dec 2025
Conestoga College *Waterloo, ON*

Google Advanced Data Analytics Professional Certificate *Jun 2026*